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Department of Computer Engineering

(Affiliated to Savitribai Phule Pune University, Pune)

Weekly Assessment Report

Used Car Price Prediction and Analysis System Using Machine Learning

A Week-wise Progress and Assessment Log Submitted in Partial
Fulfilment of the Requirements for the Course

HOCO492 — Mini Project: Artificial Intelligence and Machine Learning

Submitted by

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Weeks)

About This Report

This document records the week-wise progress of the mini project “**Used Car Price Prediction and Analysis System Using Machine Learning**” over the full **15-week** project duration (09 July 2026 – 18 October 2026), as required for continuous mentor assessment under course **HOCO492**.

Each week lists the planned objectives and the actual work performed, with references to the concrete files/artifacts produced in the project repository.

Note on structure: the technical build of this project follows the standard ML lifecycle (data acquisition → preprocessing → EDA → modelling → evaluation → deployment → testing → documentation). That lifecycle is mapped below onto the 15 available calendar weeks in the order the work was actually planned and completed.

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CHAPTER 1

Week 1: Requirement Analysis and Topic Finalization

Week	1
Duration	06 Jul 2026 – 12 Jul 2026
Objectives	Finalize the mini-project topic, define the problem statement, and align it with course outcomes CO1–CO4.

- Selected and finalized the project topic: *Used Car Price Prediction and Analysis System Using Machine Learning*.
- Defined the problem statement: used-car pricing is inconsistent and hard to estimate without domain expertise; a data-driven regression-based estimator is needed.
- Drafted project objectives and mapped them explicitly to course outcomes CO1–CO4 (independent investigation, orderly reporting, literature linkage, practical constraints).
- Reviewed the HOCO492 mini-project evaluation guidelines and scope (classical ML only, no deep learning).

Deliverable: docs/project_proposal.md (problem statement, objectives, scope, CO alignment).

CHAPTER 2

Week 2: Literature Survey

Week	2
Duration	13 Jul 2026 – 19 Jul 2026
Objectives	Survey published work on used-car / vehicle price prediction using machine learning and identify methodological gaps to address.

- Reviewed and verified five real published studies: Pudaruth (2014), Monburinon et al. (2018, ICBIR), Gegic et al. (2019, *TEM Journal*), Pal et al. (2018, FICC/Springer), and Noor & Jan (2017, *IJCA*).
- Cross-checked each paper's authors, venue, year, method, and reported result via independent search before citing.
- Synthesized a common finding across papers: tree-based ensembles consistently outperform plain linear regression for used-car pricing.
- Identified a methodological gap to avoid: several papers report only training accuracy without a comparable held-out check (motivated this project's explicit train/CV/test separation).

Deliverable: docs/literature_survey.md (5 verified references with per-paper relevance analysis).

CHAPTER 3

Week 3: Dataset Acquisition and Project Setup

Week	3
Duration	20 Jul 2026 – 26 Jul 2026
Objectives	Identify a real, public dataset suitable for the problem and set up the project’s software structure and environment.

- Located and validated the CarDekho used-car dataset (“*Car details v3.csv*”, Kaggle nehalbirla/vehicle-dataset-from-cardkho) – 8,128 rows, 13 columns, matching the required feature set.
- Set up the full project repository structure: `data/`, `src/`, `notebooks/`, `models/`, `reports/`, `tests/`, `docs/`.
- Configured the Python 3.13 environment with pandas, NumPy, scikit-learn, XGBoost, matplotlib, seaborn, Plotly, Streamlit, joblib, and pytest.
- Implemented `src/config.py` (paths, column mapping, reproducibility constants) and `src/data_loader.py` (schema-validated CSV loading).

Deliverable: `data/raw/used_cars.csv`, `src/config.py`, `src/data_loader.py`.

CHAPTER 4

Week 4: Data Cleaning and Preprocessing Pipeline

Week	4
Duration	27 Jul 2026 – 02 Aug 2026
Objectives	Clean the raw dataset and build a leakage-free preprocessing pipeline.

- Implemented `clean_raw_data()`: duplicate removal (1,202 exact duplicates found), regex-based unit stripping ("1248 CC" → 1248.0, "74 bhp" → 74.0, "23.4 kmpl" → 23.4), invalid-target row removal, and sanity filters on `km_driven/year`.
- Dropped the torque column after determining its free-text format could not be parsed reliably.
- Built the `ColumnTransformer`-based preprocessing pipeline: median imputation + standard scaling (numeric), most-frequent imputation + one-hot encoding (categorical).

Deliverable: `src/preprocessing.py`.

CHAPTER 5

Week 5: Exploratory Data Analysis and Feature Engineering

Week	5
Duration	03 Aug 2026 – 09 Aug 2026
Objectives	Explore the cleaned dataset and engineer features that improve model learnability.

- Built and executed notebooks/01_eda.ipynb covering all required EDA items: distributions, missing-value and duplicate analysis, correlation matrix, price vs. age, km driven, mileage, and engine capacity, price by fuel type, transmission, and seller type, and outlier analysis.
- Implemented car_age (current year – manufacturing year), brand extraction from the free-text name column, and rare-brand grouping (<20 listings → “Other”).
- Saved all generated EDA plots to reports/figures/.

Deliverable: notebooks/01_eda.ipynb, src/feature_engineering.py, reports/figures/*.png.

CHAPTER 6

Week 6: Baseline Model Development

Week	6
Duration	10 Aug 2026 – 16 Aug 2026
Objectives	Implement the core training pipeline and establish baseline regression performance.

- Implemented `src/train.py`: data preparation, 80/20 train–test split (`random_state=42`), and the candidate-model registry.
- Trained and evaluated baseline Linear Regression, Ridge Regression, and Decision Tree models.
- Established the baseline test $R^2 \approx 0.71$ (linear models) as the bar subsequent ensemble models must clear.

Deliverable: `src/train.py` (baseline stage), `notebooks/02_preprocessing.ipynb`.

CHAPTER 7

Week 7: Ensemble Models and Cross-Validation

Week	7
Duration	17 Aug 2026 – 23 Aug 2026
Objectives	Extend the candidate set to ensemble methods and compare all candidates via 5-fold cross-validation.

- Added Random Forest, Gradient Boosting, and XGBoost to the candidate model set (6 algorithms total).
- Implemented 5-fold cross-validation (`cross_validate_models()`) reporting CV RMSE, R^2 , and MAE (mean $\pm$ std) for every candidate on the training split only.
- Observed tree-based ensembles clearly outperforming linear models (CV $R^2 \approx 0.87$ – 0.88 vs. ≈ 0.69), consistent with the literature survey findings (Week 2).

Deliverable: `reports/cv_results.csv`.

CHAPTER 8

Week 8: Hyperparameter Tuning

Week	8
Duration	24 Aug 2026 – 30 Aug 2026
Objectives	Tune the strongest candidate models within a bounded search budget.

- Defined bounded hyperparameter search spaces for Random Forest and XGBoost (`n_estimators`, `max_depth`, `learning_rate`, `subsample`, etc.).
- Ran `RandomizedSearchCV` (20 iterations, 5-fold CV) on both models via `tune_model()`.
- Logged tuned CV RMSE and best hyperparameters for each; observed that tuning gains were modest and did not uniformly beat the untuned models on the later test-set check (Week 9).

Deliverable: Tuned model pipelines and logged best-parameter sets (see `src/train.py::tune_model()`).

CHAPTER 9

Week 9: Final Evaluation and Model Selection

Week	9
Duration	31 Aug 2026 – 06 Sep 2026
Objectives	Evaluate every candidate once on the held-out test set and objectively select the final model.

- Evaluated all 8 candidates (6 baseline + 2 tuned) exactly once on the untouched 20% test split.
- Selected **Gradient Boosting** as the final model – objectively the lowest test RMSE (Rs. 151,043, $R^2 = 0.9281$, MAPE = 21.28%), narrowly ahead of the tuned Random Forest and XGBoost.
- Verified no evidence of overfitting (test R^2 not lower than the CV R^2 range).
- Generated diagnostic plots: actual-vs-predicted, residual plot, error distribution, feature importance.
- Serialized the final bundled pipeline (preprocessing + model) with joblib.

Deliverable: `reports/model_comparison.csv`, `reports/results.md`, `models/used_car_price_model.pkl`, `models/model_metadata.json`, `reports/figures/*.png`.

CHAPTER 10

Week 10: Streamlit Application Development

Week	10
Duration	07 Sep 2026 – 13 Sep 2026
Objectives	Build the interactive web application around the trained pipeline.

- Implemented `src/predict.py`: input validation rules (year range, non-negative km, plausible mileage/engine/power bounds, known categorical values) and the prediction interface, including the empirical price-range calculation from test-set residual quantiles.
- Built `app.py` with five pages: *Home*, *Price Prediction*, *Data Analysis*, *Model Performance*, *About Project* – all reading live from generated reports/artifacts, nothing hard-coded.
- Manually verified the Price Prediction form end-to-end with sample vehicle inputs.

Deliverable: `app.py`, `src/predict.py`.

CHAPTER 11

Week 11: Automated Testing, Reliability and Containerization

Week	11
Duration	14 Sep 2026 – 20 Sep 2026
Objectives	Validate the system with automated tests and package it for portable deployment.

- Wrote 40 automated `pytest` test cases across `tests/test_data.py`, `test_preprocessing.py`, `test_model.py`, and `test_prediction.py` – all 40 passing.
- Ran reliability testing: 50 repeated predictions on identical input (0% variance, deterministic output) and measured per-prediction latency (mean 2.99 ms on a warm model).
- Authored `Dockerfile` and `docker-compose.yml`; built the image, ran the container, and confirmed the health check reports healthy and the app serves HTTP 200.

Deliverable: `tests/*.py`, `reports/test_cases.md`, `reports/reliability_test.md`, `Dockerfile`, `docker-compose.yml`.

CHAPTER 12

Week 12: Project Documentation

Week	12
Duration	21 Sep 2026 – 27 Sep 2026
Objectives	Produce the full supporting documentation set expected for the course (SDLC, design, methodology, limitations).

- Authored `docs/sdlc.md` (Agile sprint plan mapped to this project's actual phases).
- Authored `docs/system_design.md` (layered architecture, data-flow tables) and `docs/methodology.md` (full pipeline, leakage-avoidance rationale).
- Authored `docs/limitations.md` and `docs/future_scope.md`, documenting honest constraints (regional scope, dropped torque feature, empirical price range) rather than overstating capability.
- Authored `docs/academic_report.md` (18-chapter report skeleton) and the top-level `README.md`.

Deliverable: `docs/*.md`, `README.md`.

CHAPTER 13

Week 13: Project Synopsis Report (LaTeX)

Week	13
Duration	28 Sep 2026 – 04 Oct 2026
Objectives	Prepare the formal, submission-ready project synopsis document.

- Authored the full LaTeX synopsis (`reports/synopsis.tex`) following standard black-book formatting: Times 12pt, justified body text, centered chapter headings, roman/arabic page numbering, footer pagination, List of Figures/Tables.
- Captured real application screenshots via automated browser testing (Playwright driving the live Streamlit app) for the Price Prediction, Data Analysis, and Model Performance pages.
- Added system architecture diagrams (layered architecture, end-to-end ML pipeline), the mathematical formulation of every algorithm and metric used, and full training-algorithm pseudocode.
- Compiled with `MiKTeX/pdf latex`; verified via page-by-page ink-coverage audit that no page is more than half blank.

Deliverable: `reports/synopsis.pdf`.

CHAPTER 14

Week 14: Final Verification and Refinement

Week	14
Duration	05 Oct 2026 – 11 Oct 2026
Objectives	Re-verify the entire system end-to-end and polish report formatting.

- Re-ran `python -m src.train` end-to-end and confirmed all generated metrics/plots are reproducible from the included dataset with the fixed random seed.
- Re-ran the full `pytest` suite (40/40 passing) and re-verified the Docker image builds and serves correctly.
- Refined report formatting per review feedback: fully bordered tables, justified/centered layout, footer-based pagination, proper font standards.

Deliverable: Finalized `reports/synopsis.pdf`; verified reproducibility end-to-end.

CHAPTER 15

Week 15: Viva Preparation and Final Submission

Week	15
Duration	12 Oct 2026 – 18 Oct 2026
Objectives	Prepare for the viva-voce examination and complete final project submission.

- Prepared docs/viva_questions.md (44 anticipated questions with answers spanning ML fundamentals, preprocessing, evaluation metrics, and system design).
- Prepared docs/presentation_outline.md (12-slide defense presentation).
- Performed a final walkthrough of the live application and all generated reports as rehearsal for the viva demonstration.
- Assembled the complete project repository (code, trained model, tests, Docker setup, documentation, and this report) for final submission.

Deliverable: Complete, submission-ready project repository.

Overall Completion Summary and Sign-off

Milestone Completion Checklist

Week	Milestone	Status
1–2	Topic finalization and literature survey	Completed
3–4	Dataset acquisition, cleaning and preprocessing pipeline	Completed
5	Exploratory data analysis and feature engineering	Completed
6–8	Model development, cross-validation and hyper-parameter tuning	Completed
9	Final model evaluation and selection (Gradient Boosting)	Completed
10	Streamlit application development	Completed
11	Automated testing, reliability testing and containerization	Completed
12–13	Documentation and project synopsis report	Completed
14	Final verification and refinement	Completed
15	Viva preparation and final submission	Completed

Overall Remarks

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